

- Q.25 Describe different types of errors in measurement. (CO1)
- Q.26 Describe the working of a DSO. (CO4)
- Q.27 What are recording type instruments? Give one example recording type instrument. (CO2)
- Q.28 Draw the block diagram of a Digital Multimeter. (CO3)
- Q.29 Draw the block diagram of a basic CRO.
- Q.30 Explain the use of Earth Tester. (CO6)
- Q.31 Explain integration type digital voltmeter. (CO6)
- Q.32 Define distortion and list its types. (CO5)
- Q.33 Name and explain any four front panel controls of a CRO. (CO4)
- Q.34 Draw and explain block diagram of Q meter.
- Q.35 Describe the role of controlling torque in an indicating instrument. (CO2)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the construction and working of a Function generator with the help of block diagram in detail. (CO5)
- Q.37 Explain the construction and working principle of attraction type moving iron instruments with the help of diagram. (CO3)
- Q.38 Draw the circuit diagram of wheat stone bridge and explain how an unknown resistance can be measured using it. (CO6)

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Roll No.

4th Sem / Mechatronics

Subject:- Electrical & Electronic Instrumentation and Measurement

Time : 3Hrs.

M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
- Q.1 Resistances can be measured with the help of (CO3)
 a) Wattmeter b) Voltmeter
 c) Ammeter d) ohmmeter
- Q.2 The principle of Q-meter is based on (CO6)
 a) parallel resonance b) series resonance
 c) both a and b d) none of the above
- Q.3 Inductance is measured by which one of the following? (CO6)
 a) Wheat stone bridge b) Schering bridge
 c) De Sauty's bridge d) Maxwell's bridge
- Q.4 A moving iron instrument can be used for (CO3)
 a) D.C. only b) A.C. only
 c) D.C and AC both d) none of the above
- Q.5 Lissajous patterns can be used to measure (CO4)
 a) frequency b) phase different
 c) both a and b d) intensity of light
- Q.6 What is the smallest change in applied input that will indicate a detectable change in output of an instrument? (CO1)

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- a) Accuracy b) Precision
- c) Sensitivity d) Resolution

Q.7 Which device is used to measure earth resistance directly? (CO6)

- a) Thermostat
- b) Multimeter
- c) Earth tester
- d) Megger

Q.8 When the pointer of an indicating instrument comes to rest in the final deflected position. (CO2)

- a) only controlling torque acts
- b) only deflecting torque acts
- c) both deflecting and controlling torques act
- d) none of the above

Q.9 What is the primary purpose of a CRO probe? (CO4)

- a) To connect the signal source to the CRO and protect the instrument
- b) To amplify the input signal
- c) To display the signal on the screen
- d) To attenuate the input signal

Q.10 What is the purpose of damping torque in a measuring instrument? (CO2)

- a) To deflect the pointer
- b) To provide a restoring force
- c) To prevent oscillations and stabilize the pointer
- d) To measure the applied force

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Ammeter is always connected in _____ (CO3)

Q.12 In CRO, Vertical deflection plate are kept _____ (CO4)

Q.13 The scale of moving iron instrument is uniform. (True/False) (CO3)

Q.14 De Sauty's bridge is used for the measurement of _____ (CO6)

Q.15 The main specification of a CRO is its _____ (CO4)

Q.16 Q-Factor (Quality Factor) of a coil= _____ (CO6)

Q.17 Name the wave shape generated by time base circuit of a CRO. (CO4)

Q.18 A megger is defined as a device used to measure the _____ of electrical components and systems. (CO6)

Q.19 Write one application of Wave Analyser. (CO5)

Q.20 What is the function of Frequency counter? (CO5)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Why a bridge circuit has high accuracy? (CO6)

Q.22 Why precision is no guarantee to accuracy? (CO1)

Q.23 Explain the term loading effect in measurement. (CO1)

Q.24 What are different standards? (CO1)